

OLIST E-COMMERCE ANALYTICS

DAX Measures & Business Logic

Professional Technical Documentation

REPORT SCOPE	DAX calculations, business purpose and exact formulas
MODEL	public bi_fact_sales + DimDate
MEASURES	25 measures across 8 groups
PLATFORM	Microsoft Power BI / DAX

Executive Overview

DAX (Data Analysis Expressions) is the formula language used in Power BI to create dynamic calculations. In this project, DAX converts the Olist sales model into reusable KPIs for sales, customers, reviews, payments, delivery and seller performance.

Why DAX is used: measures respond to the current filter context. Therefore, a single measure can dynamically recalculate when users filter the dashboard by date, category, state, seller or other dimensions.

Measure Groups

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01 Base Measures

Each measure below includes its purpose, business use and exact DAX expression.

01 Revenue

What it does: Total product sales revenue.

Business use: Revenue KPI

```
SUM ( 'public bi_fact_sales'[price] )
```

02 Freight

What it does: Total freight/shipping value.

Business use: Logistics analysis

```
SUM ( 'public bi_fact_sales'[freight_value] )
```

03 GMV

What it does: Combines revenue and freight to calculate Gross Merchandise Value.

Business use: Overall transaction value

```
[Revenue] + [Freight]
```

04 Orders

What it does: Counts unique orders.

Business use: Order volume

```
DISTINCTCOUNT ( 'public bi_fact_sales'[order_id] )
```

05 Customers

What it does: Counts unique customers.

Business use: Customer base

```
DISTINCTCOUNT ( 'public bi_fact_sales'[customer_unique_id] )
```

06 Items Sold

What it does: Counts rows/items in the sales fact table.

Business use: Product volume

```
COUNTROWS ( 'public bi_fact_sales' )
```

07 AOV

What it does: Calculates average revenue generated per order.

Business use: Order value analysis

```
DIVIDE ( [Revenue], [Orders] )
```

08 Items per Order

What it does: Calculates the average items per order.

Business use: Basket-size analysis

```
DIVIDE ( [Items Sold], [Orders] )
```

09 Avg Rating

What it does: Calculates the average customer review score.

Business use: Customer satisfaction

```
AVERAGE ( 'public bi_fact_sales'[review_score] )
```

10 Delivered Orders

What it does: Counts delivered orders.

Business use: Delivery performance

```
CALCULATE ( [Orders], 'public bi_fact_sales'[order_status] = "delivered" )
```

11 Late Orders

What it does: Counts delivered orders marked as late.

Business use: Late-delivery monitoring

```
CALCULATE ( [Orders], 'public bi_fact_sales'[order_status] = "delivered", 'public  
bi_fact_sales'[is_late] = 1 )
```

12 On-time %

What it does: Calculates the percentage of delivered orders completed on time.

Business use: Service KPI

```
DIVIDE ( [Delivered Orders] - [Late Orders], [Delivered Orders] )
```

02 Time Intelligence Measures

Each measure below includes its purpose, business use and exact DAX expression.

01 Revenue YTD

What it does: Calculates cumulative revenue from the start of the year.

Business use: YTD performance

```
TOTALYTD ( [Revenue], DimDate[Date] )
```

02 Revenue LY

What it does: Returns revenue for the equivalent period last year.

Business use: Historical comparison

```
CALCULATE ( [Revenue], SAMEPERIODLASTYEAR ( DimDate[Date] ) )
```

03 YoY %

What it does: Calculates year-over-year revenue growth.

Business use: Growth analysis

```
DIVIDE ( [Revenue] - [Revenue LY], [Revenue LY] )
```

04 Rolling 30D Revenue

What it does: Calculates revenue for the latest 30-day period.

Business use: Recent sales trend

```
CALCULATE(  
    [Revenue],  
    DATESINPERIOD ( DimDate[Date], MAX ( DimDate[Date] ), -30, DAY )  
)
```

03 Review Sentiment Measures

Each measure below includes its purpose, business use and exact DAX expression.

01 Positive Reviews %

What it does: Percentage of non-blank reviews scored 4 or 5.

Business use: Positive sentiment

```
DIVIDE(  
    CALCULATE( COUNTROWS('public bi_fact_sales'), 'public bi_fact_sales'[review_score] >= 4 ),  
    CALCULATE( COUNTROWS('public bi_fact_sales'), NOT ISBLANK('public bi_fact_sales'[review_score]) )  
)
```

02 Negative Reviews %

What it does: Percentage of non-blank reviews scored 1 or 2.

Business use: Negative sentiment

```
DIVIDE(  
    CALCULATE( COUNTROWS('public bi_fact_sales'), 'public bi_fact_sales'[review_score] <= 2 ),  
    CALCULATE( COUNTROWS('public bi_fact_sales'), NOT ISBLANK('public bi_fact_sales'[review_score]) )  
)
```

04 Payment Measures

Each measure below includes its purpose, business use and exact DAX expression.

01 Payment Value

What it does: Calculates total recorded payment value.

Business use: Payment analysis

```
SUM ( 'public bi_fact_sales'[payment_value] )
```

02 Avg Installments

What it does: Calculates average payment installments.

Business use: Payment behavior

```
AVERAGE ( 'public bi_fact_sales'[payment_installments] )
```

05 Delivery & Logistics Measures

Each measure below includes its purpose, business use and exact DAX expression.

01 Avg Delivery Days

What it does: Calculates average delivery duration in days.

Business use: Logistics efficiency

```
AVERAGE ( 'public bi_fact_sales'[delivery_days] )
```

02 Late %

What it does: Calculates the percentage of delivered orders that were late.

Business use: Late-delivery rate

```
DIVIDE( [Late Orders], [Delivered Orders] )
```

06 Review Classification

Each measure below includes its purpose, business use and exact DAX expression.

01 Review Bucket

What it does: Classifies reviews as No Review, Positive, Neutral or Negative.

Business use: Review segmentation

```
SWITCH(
  TRUE(),
  ISBLANK('public bi_fact_sales'[review_score]), "No Review",
  'public bi_fact_sales'[review_score] >= 4, "Positive (4-5)",
  'public bi_fact_sales'[review_score] = 3, "Neutral (3)",
  "Negative (1-2)"
)
```

07 Seller Measures

Each measure below includes its purpose, business use and exact DAX expression.

01 Sellers

What it does: Counts unique sellers.

Business use: Seller network

```
DISTINCTCOUNT ( 'public bi_fact_sales'[seller_id] )
```

02 Revenue per Seller

What it does: Calculates average revenue generated per seller.

Business use: Seller productivity

```
DIVIDE ( [Revenue], [Sellers] )
```

08 Drillthrough Measure

Each measure below includes its purpose, business use and exact DAX expression.

01 Drill Title

What it does: Creates a dynamic drillthrough title based on selected category.

Business use: Interactive navigation

```
"Drillthrough: " &  
COALESCE( SELECTEDVALUE('public bi_fact_sales'[category]), "All" )
```

DAX Function Reference

The following functions are the main building blocks used by the supplied measures.

FUNCTION	PURPOSE
SUM	Adds numeric values.
DISTINCTCOUNT	Counts unique values such as orders, customers and sellers.
COUNTROWS	Counts rows in a table or filtered result.
DIVIDE	Performs safe division for ratios and percentages.
CALCULATE	Evaluates an expression under a modified filter context.
AVERAGE	Calculates the arithmetic mean.
TOTALYTD	Calculates cumulative year-to-date values.
SAMEPERIODLASTYEAR	Retrieves the equivalent period from the previous year.
DATESINPERIOD	Creates a date range for rolling-period analysis.
SWITCH / TRUE	Evaluates multiple conditions for classification.
SELECTEDVALUE	Returns the single selected value in the current context.
COALESCE	Provides a fallback value when an expression is blank.

Technical Notes

- The formulas are reproduced from the supplied DAX measure list; the expressions have not been changed.
- Time-intelligence measures use DimDate[Date].
- Review sentiment percentages exclude blank review scores from their denominator.
- Delivery calculations use the supplied delivery_days and is_late fields.
- This document separates technical logic from business purpose so the measures are understandable for both technical and business audiences.